

mem[0.40][31:0]

Address	Value
0x00000000	0x00000000
0x00000001	0x00000000
0x00000002	0x00000000
0x00000003	0x00000000
0x00000004	0x00000000
0x00000005	0x00000000
0x00000006	0x00000000
0x00000007	0x00000000
0x00000008	0x00000000
0x00000009	0x00000000
0x0000000A	0x00000000
0x0000000B	0x00000000
0x0000000C	0x00000000
0x0000000D	0x00000000
0x0000000E	0x00000000
0x0000000F	0x00000000
0x00000010	0x00000000
0x00000011	0x00000000
0x00000012	0x00000000
0x00000013	0x00000000
0x00000014	0x00000000
0x00000015	0x00000000
0x00000016	0x00000000
0x00000017	0x00000000
0x00000018	0x00000000
0x00000019	0x00000000
0x0000001A	0x00000000
0x0000001B	0x00000000
0x0000001C	0x00000000
0x0000001D	0x00000000
0x0000001E	0x00000000
0x0000001F	0x00000000
0x00000020	0x00000000
0x00000021	0x00000000
0x00000022	0x00000000
0x00000023	0x00000000
0x00000024	0x00000000
0x00000025	0x00000000
0x00000026	0x00000000
0x00000027	0x00000000
0x00000028	0x00000000
0x00000029	0x00000000
0x0000002A	0x00000000
0x0000002B	0x00000000
0x0000002C	0x00000000
0x0000002D	0x00000000
0x0000002E	0x00000000
0x0000002F	0x00000000
0x00000030	0x00000000
0x00000031	0x00000000
0x00000032	0x00000000
0x00000033	0x00000000
0x00000034	0x00000000
0x00000035	0x00000000
0x00000036	0x00000000
0x00000037	0x00000000
0x00000038	0x00000000
0x00000039	0x00000000
0x0000003A	0x00000000
0x0000003B	0x00000000
0x0000003C	0x00000000
0x0000003D	0x00000000
0x0000003E	0x00000000
0x0000003F	0x00000000
0x00000040	0x00000000
0x00000041	0x00000000
0x00000042	0x00000000
0x00000043	0x00000000
0x00000044	0x00000000
0x00000045	0x00000000
0x00000046	0x00000000
0x00000047	0x00000000
0x00000048	0x00000000
0x00000049	0x00000000
0x0000004A	0x00000000
0x0000004B	0x00000000
0x0000004C	0x00000000
0x0000004D	0x00000000
0x0000004E	0x00000000
0x0000004F	0x00000000
0x00000050	0x00000000
0x00000051	0x00000000
0x00000052	0x00000000
0x00000053	0x00000000
0x00000054	0x00000000
0x00000055	0x00000000
0x00000056	0x00000000
0x00000057	0x00000000
0x00000058	0x00000000
0x00000059	0x00000000
0x0000005A	0x00000000
0x0000005B	0x00000000
0x0000005C	0x00000000
0x0000005D	0x00000000
0x0000005E	0x00000000
0x0000005F	0x00000000
0x00000060	0x00000000
0x00000061	0x00000000
0x00000062	0x00000000
0x00000063	0x00000000
0x00000064	0x00000000
0x00000065	0x00000000
0x00000066	0x00000000
0x00000067	0x00000000
0x00000068	0x00000000
0x00000069	0x00000000
0x0000006A	0x00000000
0x0000006B	0x00000000
0x0000006C	0x00000000
0x0000006D	0x00000000
0x0000006E	0x00000000
0x0000006F	0x00000000
0x00000070	0x00000000
0x00000071	0

4294967295,X,X,
4294967295
X
X
X
X
X
X
X
X
X
X
X
X
X
X
X
X
X
X
X
X
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X
X
0
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0
0
0
0
0
0
X

```
//rom[x]=32'b imm7 _ rs2 _ rs1 _ f3 _ imm5 _ op // S-Type mem[]
rom[4] = 32'b0000000_00001_00001_010_00001_0100011; // sb mem[(x1 + 1)>>2] = x0
rom[5] = 32'b0000000_11111_00010_000_00010_0100011; // sb mem[(12 + 2)>>2] = x31
rom[6] = 32'b0000000_11111_00011_001_00011_0100011; // sh mem[(13 + 3)>>2] = x31
rom[7] = 32'b0000000_11111_00110_010_00100_0100011; // sw mem[(13 + 4)>>2] = x31
```

STORE

✓ mem[0..4][31:0]	4294967295,X,X
> 🧠 [0][31:0]	4294967295
> 🧠 [1][31:0]	X
> 🧠 [2][31:0]	X
> 🧠 [3][31:0]	X
✓ 🧠 [4][31:0]	X
🧠 [31]	X
🧠 [30]	X
🧠 [29]	X
🧠 [28]	X
🧠 [27]	X
🧠 [26]	X
🧠 [25]	X
🧠 [24]	X
🧠 [23]	X
🧠 [22]	X
🧠 [21]	X
🧠 [20]	X
🧠 [19]	X
🧠 [18]	X
🧠 [17]	X
🧠 [16]	X
🧠 [15]	0
🧠 [14]	0
🧠 [13]	0
🧠 [12]	0
🧠 [11]	0
🧠 [10]	0
🧠 [9]	0
🧠 [8]	0
🧠 [7]	0
🧠 [6]	0
🧠 [5]	0
🧠 [4]	0
🧠 [3]	0
🧠 [2]	0
🧠 [1]	0
🧠 [0]	0
> 🧠 [5][31:0]	1073741824
> 🧠 [6][31:0]	X
> 🧠 [7][31:0]	X



```
//rom[x]=32'b imm7 _ rs2 _ rs1 _ f3_ imm5_ op // S-Type mem[]
rom[4] = 32'b0000000_00001_00001_010_00001_0100011; // sb mem[(x1 + 1)>>2] = x0
rom[5] = 32'b0000000_11111_00010_000_00010_0100011; // sb mem[(12 + 2)>>2] = x31
rom[6] = 32'b0000000_11111_00011_001_00011_0100011; // sh mem[(13 + 3)>>2] = x31
rom[7] = 32'b0000000_11111_00110_010_00100_0100011; // sw mem[(13 + 4)>>2] = x31
```

STORE

✓	mem[0:40][31:0]	4294967295,X,X
>	0[31:0]	4294967295
>	1[31:0]	X
>	2[31:0]	X
>	3[31:0]	X
✓	4[31:0]	X
	5[31]	X
	6[30]	X
	7[29]	X
	8[28]	X
	9[27]	X
	10[26]	X
	11[25]	X
	12[24]	X
	13[23]	X
	14[22]	X
	15[21]	X
	16[20]	X
	17[19]	X
	18[18]	X
	19[17]	X
	20[16]	X
	21[15]	0
	22[14]	0
	23[13]	0
	24[12]	0
	25[11]	0
	26[10]	0
	27[9]	0
	28[8]	0
	29[7]	0
	30[6]	0
	31[5]	0
	32[4]	0
	33[3]	0
	34[2]	0
	35[1]	0
	36[0]	0
>	37[5][31:0]	1073741824
>	38[6][31:0]	X
>	39[7][31:0]	X

[illegible]

```
//rom[x]=32'b imm7 _ rs2 _ rs1 _ f3_ imm5_ op // S-Type mem[]
rom[4] = 32'b0000000_00001_00001_010_00001_0100011; // sb mem[(x1 + 1)>>2] = x0
rom[5] = 32'b0000000_11111_00010_000_00010_0100011; // sb mem[(12 + 2)>>2] = x31
rom[6] = 32'b0000000_11111_00011_001_00011_0100011; // sh mem[(13 + 3)>>2] = x31
rom[7] = 32'b0000000_11111_00110_010_00100_0100011; // sw mem[(13 + 4)>>2] = x31
```

immediate

mem[0.31][31.0]	4294967295,429
> [0][31.0]	-1
> [1][31.0]	-1
> [2][31.0]	12
> [3][31.0]	13
> [4][31.0]	-1
> [5][31.0]	12
> [6][31.0]	16
> [7][31.0]	17
> [8][31.0]	65535
> [9][31.0]	6
> [10][31.0]	20
> [11][31.0]	21
> [12][31.0]	22
> [13][31.0]	23
> [14][31.0]	24
> [15][31.0]	25
> [16][31.0]	26
> [17][31.0]	27
> [18][31.0]	28
> [19][31.0]	29
> [20][31.0]	30
> [21][31.0]	31
> [22][31.0]	32
> [23][31.0]	33
> [24][31.0]	34
> [25][31.0]	35
> [26][31.0]	36
> [27][31.0]	37
> [28][31.0]	38
> [29][31.0]	39
> [30][31.0]	40
> [31][31.0]	1073741824

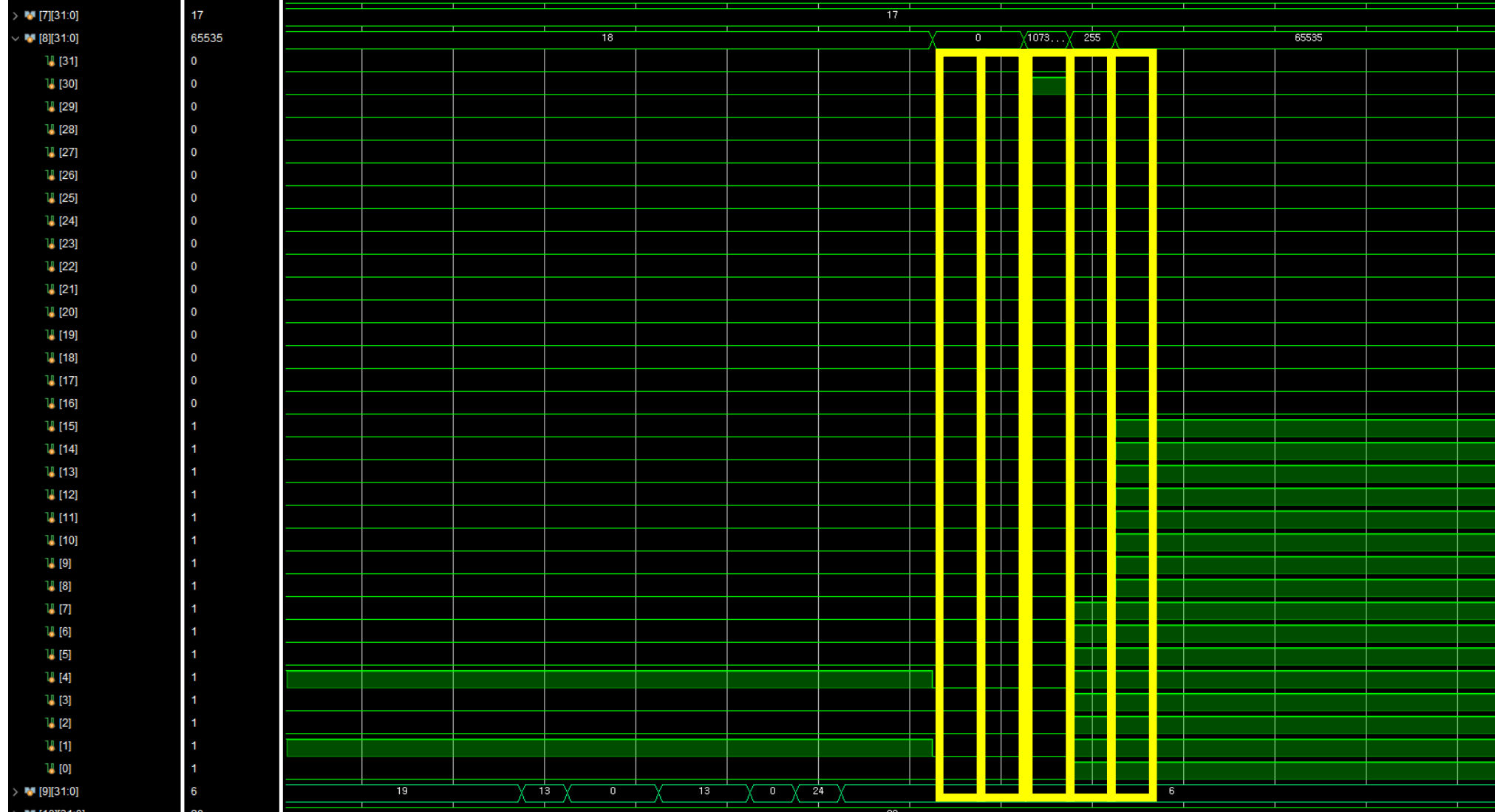
[illegible]

```
//rom[x]=32'b      imm12      _rs1_ f3_ rd      _      op      // I-Type
//or      =32'b func7 _shamt_ rs1 _ f3_ rd      _      op

rom[8] = 32'b0000000000001_00010_000_01001_0010011;// addi x9, x2, 1
rom[9] = 32'b0000000000001_00010_010_01001_0010011;// slti x9, x2, 1
rom[10] = 32'b0000000000001_00010_011_01001_0010011;// sltiu x9, x2, 1
rom[11] = 32'b0000000000001_00010_100_01001_0010011;// xori x9, x2, 1
rom[12] = 32'b0000000000001_00010_110_01001_0010011;// ori x9, x2, 1
rom[13] = 32'b0000000000001_00010_111_01001_0010011;// andi x9, x2, 1
rom[14] = 32'b00000000_00001_00010_001_01001_0010011;// slli x9, x2, 1
rom[15] = 32'b00000000_00001_00010_101_01001_0010011;// srli x9, x2, 1
rom[16] = 32'b0100000_00001_00010_101_01001_0010011;// srai x9, x2, 1
```

```
x9(13) = x2(12) + 1
x9(0) = x2(12) < 1 ==> FALSE
x9(0) = x2(12) < unsigned(1) ==> FALSE
x9(13) = x2(12) ^ 1 ==> 13
x9(13) = x2(12) | 1 ==> 13
x9(0) = x2(12) & 1
x9(24) = x2(12) << 1
x9(6) = x2(12) >> 1
x9(6) = x2(12) >> 1
```

LOAD



17 18 19 20 21

```
//rom[x]= 32'b    imm12 _ rs1 _ f3_ rd _ op  //  L-Type
rom[17] = 32'b000000010100_00000_000_01000_0000011;// lb x8 = mem[5][0:7]
rom[18] = 32'b000000010100_00000_001_01000_0000011;// lh x8 = mem[5][0:15]
rom[19] = 32'b000000010100_00000_010_01000_0000011;// lw x8 = mem[5][0:31]
rom[20] = 32'b0000000000000_00000_100_01000_0000011;// lbu x8 = mem[0][0:7]
rom[21] = 32'b0000000000000_00000_101_01000_0000011;// lhu x8 = mem[0][0:15]
```

Mem[0]=-1
Mem[5]=2**30